

Statistical Analysis Report – Use Case Diagram Compilation

Hypotheses

H_0 (Null Hypothesis): The proportion of diagrams that compile correctly is equal between the models ($p_1 = p_2$).

H_a (Alternative Hypothesis): There is a difference in the proportion of diagrams that compile between the models ($p_1 \neq p_2$).

Descriptive Statistics

Total samples: 2000

Compiled diagrams: 1519

Non-compiled diagrams: 481

Model	Compiled	Total	Proportion
Fin-R1-c1-latest	680	1000	0.6800
Qwen-Coder-c1-latest	839	1000	0.8390

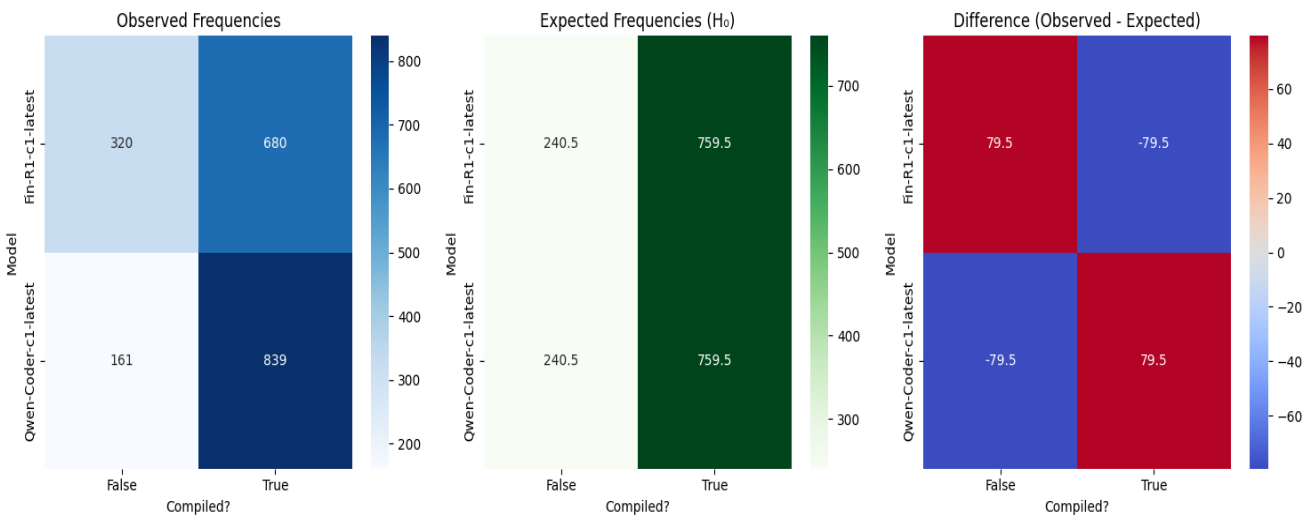
Z-Test Result

Z statistic: -8.3188

p-value: 8.8858e-17

Conclusion: Statistically significant difference found between models ($p < 0.05$).

Visual Analysis of Frequencies



Final Interpretation

The non-specialized model (Qwen-Coder-c1-latest) compiled more diagrams than the specialized one (Fin-R1-c1-latest).
This suggests a negative impact of specialization.